

Applied Human Computer Interaction (AHCI)
Course Project
Spring 2026

Sections: BCS-6H & BCS-6J

Project Description: It will be a group project with a maximum of 4 students. (Not more than 4 students). This project will assess the practical implementation of learning which we have studied in the course. Following is a brief description.

1. Each group must select any mobile / PC / web game
2. Identify usability problems
3. Apply HCI principles
4. Design improved UI
5. Create prototype
6. Compare before vs after

Functional Requirements: Project must have redesigned at least 4 screens

1. Main menu
2. Gameplay HUD (This will have multiple sub-screens like different levels)
3. Settings menu
4. Result screen

Design Requirements: Group must have ensure below areas in their design.

1. Use consistent colors
2. Apply visual hierarchy
3. Improve readability
4. Reduce cognitive load
5. Improve navigation
6. Apply accessibility

Application of HCI Principles: Students must demonstrate below in their project. There could be more principles as well other than mentioned below. Team will get extra marks if they will be able to accommodate more HCI principles in the project.

1. Usability & Accessibility
2. Consistency
3. Feedback & System Status
4. Error prevention (If applicable)
5. Cognitive load reduction
6. Enable user control/freedom

Group Tasks:

1. **Group making:** Maximum of 4 group members in each group. No group should have more than 4 students. It is suggested to have at least 3 members in each group.
2. **Game Selection:** Each group must select mobile / PC / web game. (Candy Crush, Ludo Star, Subway Surfers)
3. **Usability Problems:** Identify at least 5 problems like Example: cluttered UI, small buttons, confusing navigation, poor feedback)
4. **Fidelity Design:** Create improved UI. Any tool can be used to create improved UI. Some tools are Figma, Canva etc.
5. **Usability Evaluation:** Compare Old UI vs New UI
Task Completion time
Error rate
User Satisfaction

Deliverables & Submissions:

1. **Group Making & Game Selection:** Each group should submit a single entry that includes the names of all team members and their chosen game. The submission should also briefly explain why the game was selected and highlight potential areas for improvement. Submission deadline would be **Saturday, 18th April 2026 11:59 pm.**
2. **Final Submission:** Final submission will be due on **Saturday, 2nd May 2026 11:59 pm.** which must have Project Report (PDF format), Final UI screens, usability evaluation and final presentation (PPT format).
 - **Project Report:** It should include **Project report** (PDF format) explaining whole project working.
 - **Final UI screens:** It should include comparison of before vs after comparison. Your work should be clearly visible, and differences can be visualized properly.
 - **Usability evaluation:** Group needs to submit usability evaluation mentioning all the attributes with old and new comparison.
 - **Final Presentation:** It should include presentation (PPT format) having all relevant details which will be presented in class by each group.

AHCI Project Marking Rubric

Marks	Criteria
1. Game Selection & Problem Justification (10 Marks)	
9–10	Game selected appropriately, clear justification, strong improvement areas identified
7–8	Good game selection with reasonable justification

5-6	Game selected but weak reasoning
3-4	Poor justification, unclear problem
0-2	No proper selection or explanation
2. Usability Problems Identification (10 Marks)	
9-10	5+ clearly explained usability problems with examples.
7-8	4 usability problems identified clearly
5-6	3 problems identified
3-4	1-2 problems unclear
0-2	No proper usability analysis
3. Application of HCI Principles (10 Marks)	
9-10	5+ HCI principles correctly applied
7-8	4 principles applied
5-6	2-3 principles applied
3-4	Minimal application
0-2	No HCI principles used
4. UI Redesign (10 Marks)	
9-10	4+ screens (multiple sub screens) with strong improvement of UI, consistent design
7-8	4+ screens (multiple sub screens) with good UI design
5-6	4 screens with limited improvement
3-4	3-4 screens with incomplete or weak UI
0-2	Missing screens
5. Usability Evaluation (10 Marks)	
9-10	Proper comparison with metrics table and explanation
7-8	Comparison present with some metrics
5-6	Basic comparison only
3-4	Weak evaluation
0-2	No Comparison
6. PDF Report (10 Marks)	
9-10	Complete report, well structured, clear visuals
7-8	Good report, minor missing sections
5-6	Average report
3-4	Incomplete report
0-2	Poor report
7. Final Presentation (20 Marks) Each student will be evaluated individually. Instructor will ask questions from each group member, and responses will be included in grading.	
17-20	Excellent explanation of project, strong HCI understanding, professional slides, active participation, answers confidently and correctly
13-16	Average explanation, basic understanding of project, limited participation, answers some questions
9-12	Average explanation, basic understanding of project, limited participation, answers some questions
5-8	Weak explanation, poor understanding of HCI concepts, minimal participation, unable to answer most questions
0-4	Poor presentation, cannot explain project, no participation, unable to answer questions